Plasma: It is slightly alkaline, containing non-cellular substance which constitutes about 55% of the blood. Organic substances like proteins, glucose, urea, enzymes, hormones, vitamins and minerals are present in the plasma.

Formed Elements of Blood: Blood corpuscles are of three types

1. **Red blood corpuscles** (RBC) or **Erythrocytes**
2. **White blood corpuscles** (WBC) or **Leucocytes**
3. **Blood platelets** or **Thrombocytes**.

Red blood corpuscles (Erythrocytes)



Erythrocytes

They are the most abundant cells in the human body. RBCs are formed in the bone marrow. The RBCs impart red colour to the blood due to presence of respiratory pigment **haemoglobin.** Matured mammalian RBCs do not have cell organelles and nucleus. They are biconcave and disc-shaped. Their life span is about 120 days. RBC is involved in the transport of oxygen from lungs to tissues.



Why does mammalian RBC lack cell organelles and nucleus?

Mammalian RBC lack nucleus and makes the cells biconcave and increase surface area for oxygen binding, loss of mitochondria allows the RBC to transport all the oxygen to tissues, and loss of endoplasmic reticulum allows more flexibility for RBC to move through the narrow capillaries.

White blood corpuscles (Leucocytes)

WBC's are colourless. They do not have haemoglobin and are nucleated cells. It is found in the bone marrow, spleen, thymus and lymph nodes. They are capable of amoeboid movement

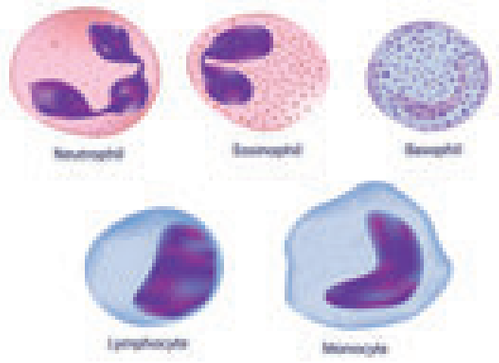


Figure 14.9 Leucocytes

The white blood corpuscles can be grouped into two categories:

1. Granulocytes 2. Agranulocytes.

Granulocytes

They contain granules in their cytoplasm. Their nucleus is irregular or lobed. The granulocytes are of three types

(i) Neutrophils (ii) Eosinophils
(iii) Basophils

(i) Neutrophils

They are large in size and have a 2 - 7 lobed nucleus. These corpuscles form 60% - 65% of the total leucocytes. Their numbers are increased during **infection** and **inflammation**.

(ii) Eosinophils

It has a bilobed nucleus and constitute 2% - 3% of the total leucocytes. Their number increases during conditions of **allergy** and **parasitic infections.** It brings about detoxification of toxins.

(iii) Basophils

Basophils have lobed nucleus. They form 0.5-1.0% of the total leucocytes. They release chemicals during the process of **inflammation.**

Agranulocytes

Granules are not found in the cytoplasm of these cells. The agranulocytes are of two types:

(i) Lymphocytes (ii) Monocytes

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